Carrie (Jiayi) Lei-Cramer

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Research interests: scalable Gaussian processes; conformal prediction and uncertainty quantification; spatially varying coefficient models, scientific machine learning.

Education

Texas A&M University Aug. 2022 – May 2027 (expected)
Ph.D. in **Statistics** (Advisors: Dr. Huiyan Sang & Dr. Matthias Katzfuss) College Station, TX

University of California, Davis 2019
B.S. in **Applied Mathematics**; Minor in **Computer Science** Davis, CA

Research Experience

Graduate Research Assistant Aug. 2023 – Present
Texas A&M University College Station, TX

– Developing deep-learning methods for sparse spatially varying coefficient models, enabling variable selection and coefficient estimation. Manuscript in preparation [P3]. Advisor: Dr. Huiyan Sang.

Research Scholar and Research Intern May 2025 – May 2026
Lawrence Livermore National Laboratory Livermore, CA

– Developed scalable adaptive conformal prediction methods for uncertainty quantification of an inertial confinement fusion (ICF) emulator, reducing prediction-interval width by up to 35% while maintaining target coverage. Selected as a Top 3 Research Project (of 30); first-author manuscript [P2]. Mentor: Dr. Laura Wendelberger.

– Extended the Diffusion-VAS model with temporal and geometric refinements, improving amodal video object segmentation by 10% mIoU.

Research Intern May 2024 – Aug. 2024
NASA Jet Propulsion Laboratory Remote

– Extended non-Gaussian generative models to enable robust forward prediction from incomplete trajectory data. Mentor: Dr. Jon Hobbs.

Graduate Research Assistant Aug. 2022 – Aug. 2023
Texas A&M University College Station, TX

– Developed scalable generative models for non-Gaussian spatio-temporal fields via autoregressive Gaussian processes; first-author manuscript [P1]. Advisor: Dr. Matthias Katzfuss.

Undergraduate Researcher Jan. 2019 – Jun. 2019
University of California, Davis Davis, CA

– Proposed and implemented a manifold proximal gradient method for non-negative principal component analysis. Advisor: Dr. Shiqian Ma.

Publications and Manuscripts

[P1] **C. Lei-Cramer**, J. Cao, M. Katzfuss. "Scalable Stochastic Modeling of Non-Gaussian Spatio-Temporal Fields via Autoregressive Gaussian Processes." *Revision submitted, Advances in Statistical Climatology, Meteorology and Oceanography.* arXiv:2605.03152.

[P2] **C. Lei-Cramer**, M. Jones, L. Wendelberger. "Scalable Adaptive Conformal Prediction with Application to an Inertial Confinement Fusion Emulator." *Submitted.* arXiv:2610.00535.

[P3] **C. Lei-Cramer**, H. Sang. "Deep Learning for Sparse Spatially Varying Coefficient Models." *In preparation.*

Fellowships, Honors, and Awards

- DOE-NNSA Office of Defense Nuclear Nonproliferation (DNN) TecH Consortium Fellowship 2026
- Top 3 Research Project (of 30), Lawrence Livermore National Laboratory 2025
- NSF Graduate Research Fellowship Program (GRFP), Honorable Mention 2024
- Outstanding Performance Award, Department of Mathematics, UC Davis 2019

Presentations

Generative Modeling for Non-Gaussian Spatio-Temporal Fields (poster) Aug. 2023
Joint Statistical Meetings (JSM) Toronto, Canada

Generative Modeling for Non-Gaussian Spatio-Temporal Fields (poster) Jul. 2023
Spatial Statistics Conference Boulder, CO

Professional Service

Session Chair, *Recent Advances in Uncertainty Quantification for Complex Systems* Aug. 2023
Joint Statistical Meetings (JSM) Toronto, Canada

Teaching Experience

Graduate Teaching Assistant Aug. 2023 – Present
Texas A&M University College Station, TX

- Statistical Computing; Mathematical Statistics; Spatial Statistics; Applied Statistics.

Lecture Note Editor Jul. 2019 – Oct. 2019
Department of Mathematics, UC Davis Davis, CA

- Proofread and typeset lecture notes for *Mathematics for Data Analytics and Decision Making.*

Industry Experience

Data Scientist Intern Jun. 2026 – Aug. 2026
American Express New York, NY

- Trained gradient-boosted models for acquisition profitability on 100M+ records, improving rank-order discrimination by 4 Gini points over the production baseline.
- Built a reusable model-diagnostic harness (feature importance, segment-level error, sensitivity analysis), cutting candidate-model iteration time from 12 hours to 5 minutes.

Senior Marketing Data Analyst Feb. 2020 – Aug. 2022
AAA Walnut Creek, CA

- Automated lead and sales classification in Python (keyword detection, decision trees), saving 20 team-hours per week.
- Identified drivers of customer satisfaction with hypothesis testing and regression in R, informing operations training and performance goals.

Skills

Methods	Gaussian processes, conformal prediction, UQ, Bayesian inference, diffusion/generative models, spatial statistics
Programming	Python (PyTorch), R, C++, MATLAB
Computing	Slurm, SQL, Google BigQuery, Git, LaTeX